

30-5-2016(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam
April - May 2016

Max. Marks: 100

Class: SE

Name of the Course: Object Oriented Programming Methodology

Course Code: CSC302

Duration: 03Hrs

Semester: III

Branch: Computer Engg.

MUCBGS**Instructions:**

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks								
Q 1 (a)	Explain JVM architecture.	05								
Q 1 (b)	Differentiate between String and String buffer class.	05								
Q1(c)	Write a program to check whether a number is palindrome or not. Example:12321	10								
Q2 (a)	The results of 5 students are tabulated as follows <table><tr><th>RollNo</th><th>Subject1</th><th>Subject2</th><th>Subject3</th></tr><tr><td></td><td></td><td></td><td></td></tr></table> Write the program (using 2-D array) to read the data and determine the following: a) Total marks obtained by each student b) The highest marks in each subject and the RollNo of student who secured it. c) The student who obtained highest total marks OR What is Vector class? Explain methods of vector class with example	RollNo	Subject1	Subject2	Subject3					10
RollNo	Subject1	Subject2	Subject3							
Q.2 (b)	What are constructors? Explain different types of constructors with example.	10								
Q3 (a)	Differentiate between Application and Applet. Write a applet program to display Smiley face.	10								
Q3 (b)	Explain with a neat diagram life cycle of thread and Explain any five methods of thread class. OR Write a program to display 'HELLO BYE HELLO BYE HELLO BYE' using multiple threads.(one thread to display HELLO and other thread to	10								

	display BYE)	
Q4 (a)	Explain Exception handling feature of Java with suitable example OR Define a user defined exception called "VotingException" that is thrown when a user age is not greater than 18. Write a program with use of this exception.	10 10
Q4 (b)	Draw class Diagram for the Railway Reservation System. Clearly show attributes, multiplicities and aggregations/compositions/Association between classes in the class diagram.	10
Q5 (a)	A class 'Array' stores an array reference, capacity of array and number of elements currently stored. It also contains a parameterized constructor and display function. Derive class Array1 which adds a function insert to insert a new value at the end of the last element (without checking for overflow). A class 'Array2' is derived from 'Array1' which overrides the insert functions. The new insert function first checks for the overflow and inserts the element only if the array is not full. Write a program using above class hierarchy which also provides the following programming element. a) use dynamic dispatch b) the program should not allow creation of objects of class Array1 and Array OR Write short note on: Access protection in java Write a short note on: Wrapper classes in JAVA	10 05 05
Q5(b)	1) Explain System.arraycopy () with example. 2) Write a Difference between Array and Vector.	05 05